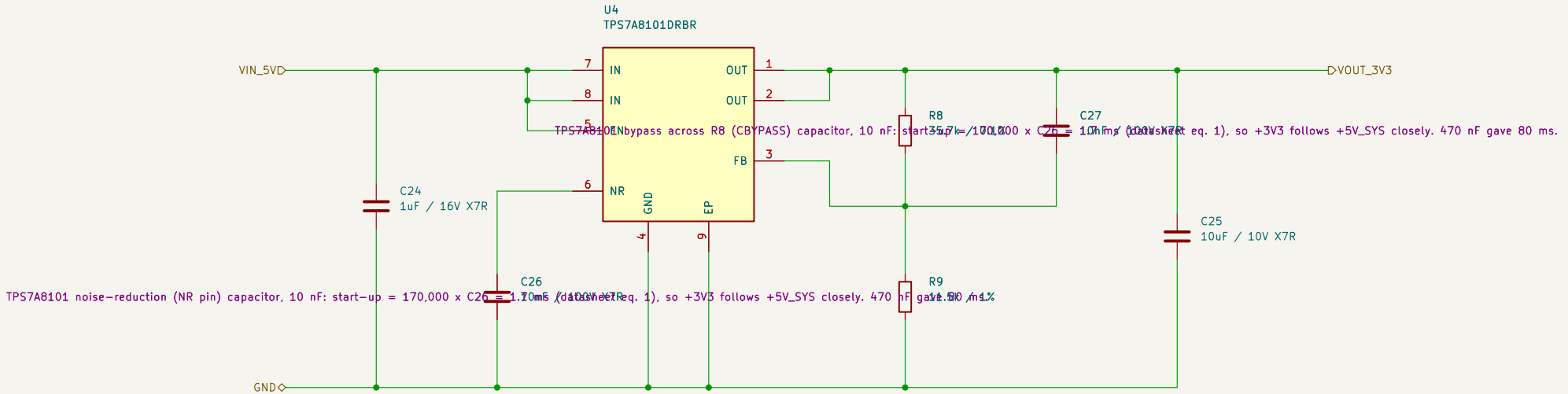
+3V3
C1 100nF
C2 100nF
C3 100nF
C4 100nF
C19 100nF
C20 4.7uF
C21 4.7uF
C22 1uF
C23 100nF
600R FB2
+3.3VA
GND
these two caps for VDDA pin 13
+3V3
R25 10k
NRST
C13 100nF
RESET SW1
GND
GND
+3V3
BOOT
J4 BOOT0 jumper
Jumper open = run from flash
Jumper fitted + reset = ROM bootloader
GND
Y3 8MHz
OSC_IN
OSC_OUT
GND
C5 16pF
C6 16pF
GND
GND
+3V3
SWCLK
SWDIO
NRST
GND
J1 SWD
programming header
Power Indicators
File: Power_Indicators.kicad_sch
Test Points
File: Test_Points.kicad_sch
Motor Control
MOTOR1_PWM
MOTOR1_DIR
ENC1_A
ENC1_B
MOTOR2_PWM
MOTOR2_DIR
ENC2_A
ENC2_B
File: Motor_Control.kicad_sch
M1: PWM PA6, DIR PB0, ENC PA15/PB3 (TIM3/TIM2)
M2: PWM PA7, DIR PB2, ENC PA8/PA9 (TIM3/TIM1)
MCU GPIO Breakout
File: GPIO_Breakout.kicad_sch
J8–J10: 2.54 mm headers for every spare STM32 pin.
Connected by global labels, so the block has no sheet pins.
+3V3
+3.3VA
+3V3
NRST
OSC_IN
OSC_OUT
BOOT
PC0
PC2
PC6
PC7
PC10
PC11
U1
STM32F091RCT6
PA0 VBAT_SENSE
PA1
PA2 USART2_TX
PA3 USART2_RX
PA4
PA5 LED_USER
PA6 MOTOR1_PWM
PA7 MOTOR2_PWM
PA8 ENC2_A
PA9 ENC2_B
PA10
PA11
PA12
PA13 SWDIO
PA14 SWCLK
PA15 ENC1_A
PB0 MOTOR1_DIR
PB2 MOTOR2_DIR
PB3 ENC1_B
PB6
PB7
PB9
PB10 SERVO_TX
PB11 SERVO_DIR
PB12 LED_PI_LINK
PB13
PB14 LED_MOTOR
PB15
VSS VSSA
GND
Mounting holes – 4x M3, 3.2 mm non-plated (NPTH), no copper, no net.
Place one per board corner; not in the BOM (no part to buy).
H1 M3
H2 M3
H3 M3
H4 M3
H5 M3
VBAT_FUSED
J2 XT30PW-M
VBAT_IN
F1 10A / 32VDC
+12V
C36 220uF 35V
GND
GND
Input protection (J2 battery) – see the Input Protection sheet
J2 XT30 -> VBAT_IN -> F1 10 A MINI fuse -> VBAT_FUSED -> Q1 reverse-battery P-FET -> +12V (D13 SMBJ16A TVS, C36 bulk, battery sense on PA0).
C36 is on +12V after Q1, so it is never reverse-biased.
Battery harness fuse at the M8 lugs: 15 A.
Input Protection
File: Input_Protection.kicad_sch
Q1 reverse-battery P-FET, D13 TVS, VBAT_SENSE (PA0)
Pi 5 Interface
USART2_TX
USART2_RX
+5V_PI
USART2_TX
USART2_RX
VIN_5V
File: Pi5_Interface.kicad_sch
J13 Pi link, 6-pin JST-XH: UART (R12/R13 1k), GND, Pi 3V3 (enables AUX),
STM32 reset (Q2) and BOOT0 for reflashing over USART2
J14 Pi power (+5V_PI direct, no fuse)
AUX Outputs
File: AUX_Outputs.kicad_sch
J18–J21: four 5 V peripheral ports, one PTC each
Servo Interface
File: Servo_Interface.kicad_sch
J17 latch servo: fused VBAT, U8 half-duplex UART (PB10/PB11)
Buck 5V SYS
+12V
VIN_12V
VOUT_5V
+5V_SYS
GND
GND
File: Buck_5V_2A.kicad_sch
12 V to 5 V / 2 A buck – U3 (TPS54202)
Buck 5V PI
+12V
VIN_12V
VOUT_5V
+5V_PI
GND
GND
File: Buck_5V_6A.kicad_sch
Buck 5V AUX
+12V
VIN_12V
VOUT_5V
+5V_AUX
GND
GND
File: Buck_5V_6A_AUX.kicad_sch
12 V to 5 V / 6 A buck – U2 (TPSM63606)
Output is the +5V_PI net (feeds the Pi via J14)
LDO Power
+5V_SYS
VIN_5V
VOUT_3V3
+3V3
GND
GND
File: LDO_3V3.kicad_sch
3.3 V / 1 A LDO – U4
STM32 VDD and analog filter input supplied from +3V3
ECE49022 Team 22 senior design – STM32F091 control, power and I/O
Derived from ST NUCLEO-64 MB1136 rev C-04, sheets 2-3
Sheet: /
File: delivery_robot_1.kicad_sch
Title: Delivery Robot Main Board
Size: A3
Date: 2026-10-08
Rev: 1.0
KiCad E.D.A. 10.0.6
Id: 1/13

# 5 V to 3.3 V LDO supply

Reusable sheet: VIN_5V input, VOUT_3V3 output, common GND




VOUT nominal = 0.8 x (1 + 35.7k / 11.5k) = 3.284 V (E96 values)
1 A is the TOTAL output budget for the STM32 and other 3.3 V loads.

C25: require >= 4.7 uF EFFECTIVE capacitance; selected part modeled at 8.8 uF.
EN tied to IN. C26 (NR) and C27 (bypass across R8) are 10 nF: start-up about 1.7 ms
(TI tests the part with CNR = 10 nF, CBYPASS = 0). 470 nF gave 80 ms, during which
5 V encoder and servo signals could reach MCU pins while VDD was still 0.

Layout: capacitors close to U4; EP pad 9 to GND copper and thermal vias.
Heat: the real 3.3 V load is about 150 mA, about 0.25 W at 5 V input
(1.7 W only at the full 1 A rating).

5 V motor / encoder supply branches BEFORE this sheet.

5 V to 3.3 V, 1 A total rail capacity

**Senior Design**

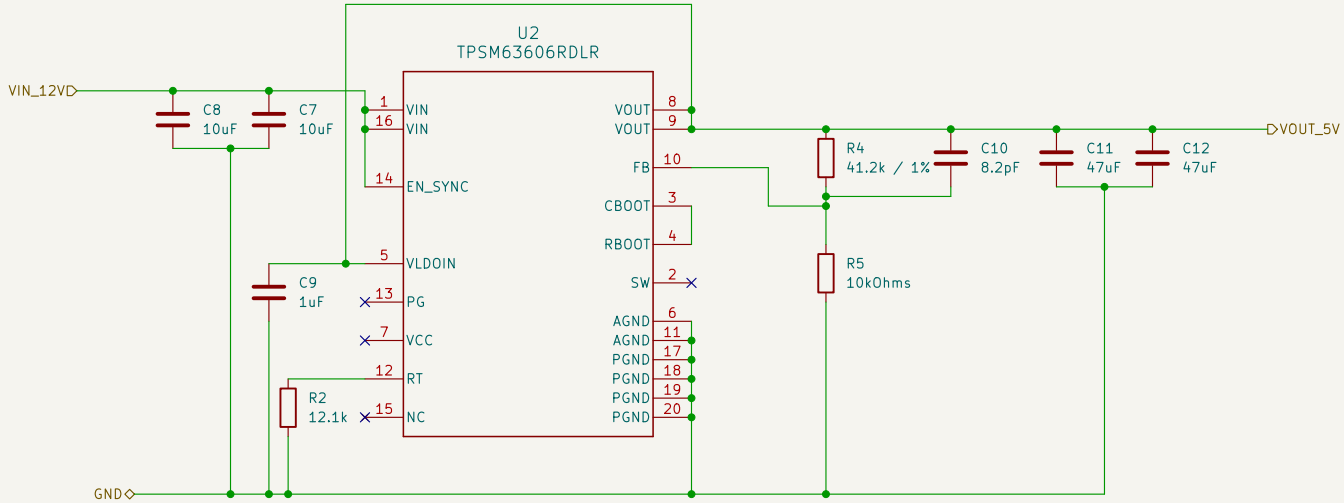
Sheet: /LDO Power/
File: LDO_3V3.kicad_sch

***Title: LDO power supply — reusable sheet***

| Size: A4 | Date: 2026-10-05 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/13 |

# 12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U2)




Reusable sheet: VIN_12V input, VOUT_5V output, common GND
VOUT set by R4/R5 divider on FB; RT = R2; EN tied to VIN.
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U2), 6 A rail

**Senior Design**

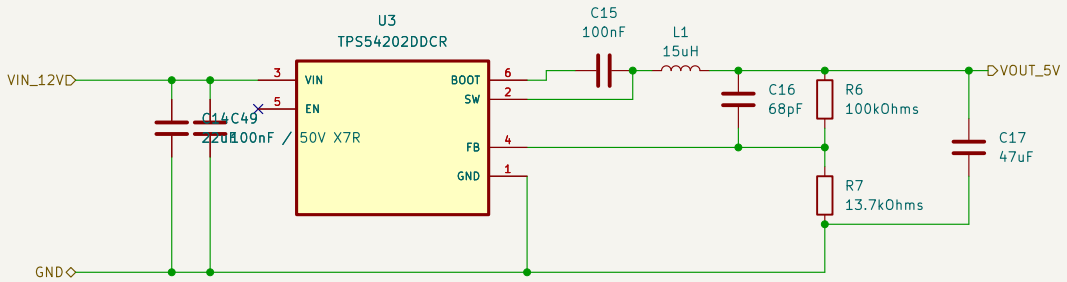
Sheet: /Buck 5V PI/
File: Buck_5V_6A.kicad_sch

***Title: Buck converter – reusable sheet***

| Size: A4 | Date: 2026-10-06 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/13 |

# 12 V – 5 V, 2A Buck Convertor

12 V to 5 V buck converter – TPS54202 (U3)




Reusable sheet: VIN_12V input, VOUT_5V output, common GND

12 V to 5 V buck, TPS54202 (U3), 2 A rail

**Senior Design**

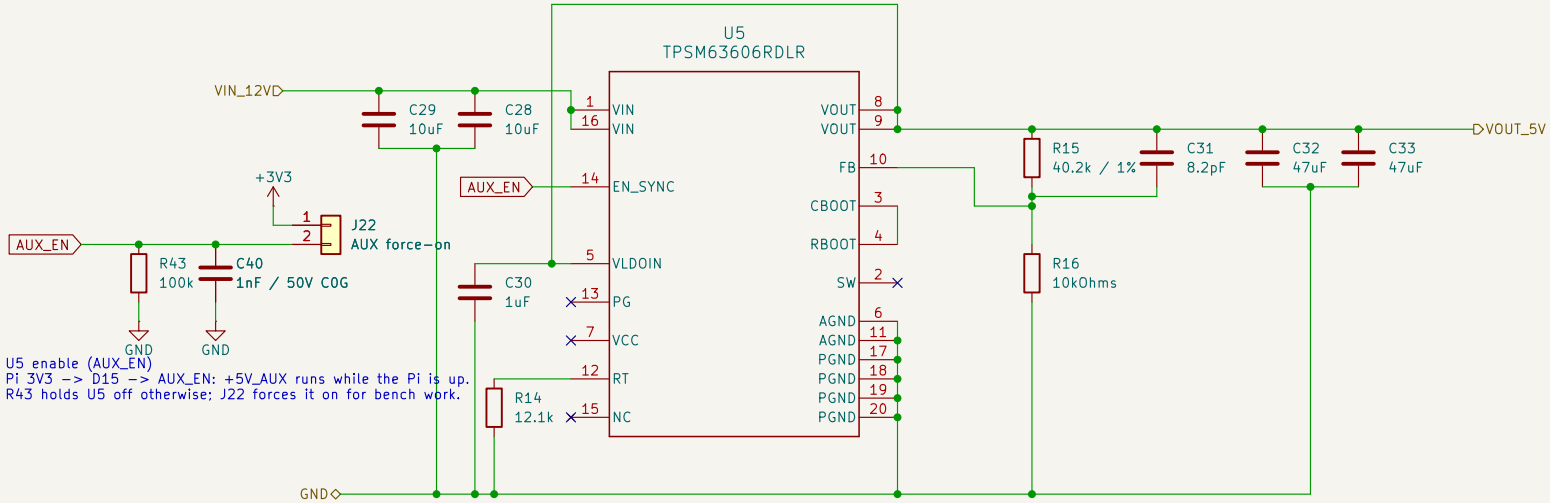
Sheet: /Buck 5V SYS/
File: Buck_5V_2A.kicad_sch

***Title: Buck converter – reusable sheet***

| Size: A4 | Date: 2026-10-06 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/13 |

# 12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U5)




Reusable sheet: VIN_12V input, VOUT_5V output, common GND
VOUT set by R15/R16 divider on FB; RT = R14; EN = AUX_EN (Pi-controlled, see note at left).
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U5), 6 A rail

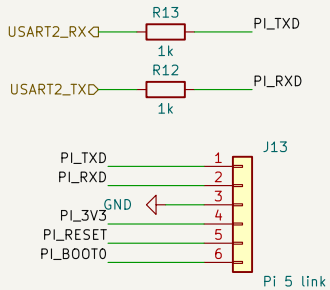
**Senior Design**

Sheet: /Buck 5V AUX/
File: Buck_5V_6A_AUX.kicad_sch

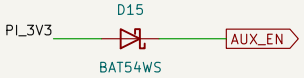
***Title: Buck converter – reusable sheet***

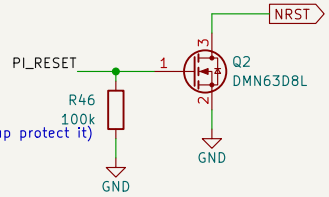
| Size: A4 | Date: 2026-10-06 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/13 |

Raspberry Pi 5 link (USART2, PA2 TX / PA3 RX), Pi 3V3 sense, STM32 reset and BOOT0

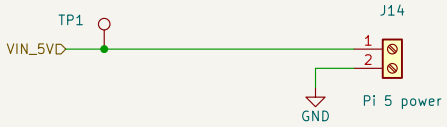



J13 1 <- Pi GPIO14 TXD
J13 2 -> Pi GPIO15 RXD
J13 3 = GND (Pi header pin 6); 3.3 V logic
R12/R13 near U1 (no ESD array fitted)




Pi 5 power: +5V_PI -> J14 USB-C pigtail (no fuse: U2's current limit and hiccup protect it)




Bring-up: leave J14 unwired, check TP1; load-test J14;
then connect the Pi. 5-6 A path




Pi-controlled functions (J13 4 <- Pi 3V3 pin 1, 5 <- GPIO23 pin 16, 6 <- GPIO24 pin 18)
AUX_EN: the +5V_AUX buck (U5) runs only while the Pi's 3V3 is up.
PI_RESET + PI_BOOT0 -> reflash the STM32 from the Pi over USART2 (AN2606:
PA2/PA3 is a bootloader port on STM32F09x). Sequence: BOOT0 high, pulse reset,
run stm32flash at 115200 8E1, then BOOT0 low and reset.
At Pi boot GPIO23/24 default to pull-down inputs, so the STM32 runs normally.

**Senior Design**

Sheet: /Pi 5 Interface/
File: Pi5_Interface.kicad_sch

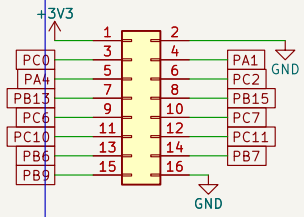
***Title: Raspberry Pi 5 interface — UART and power***

| Size: A4 | Date: 2026-10-07 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 6/13 |

MCU expansion header J8 — 2.54 mm male pins, spare MCU pins only
SPI2: PB13 SCK, PC2 MISO, PB15 MOSI, PB9 CS  |  I2C1: PB6 SCL, PB7 SDA (no pull-ups on board)
USART3: PC10 TX, PC11 RX  |  ADC: PC0, PA1, PA4 (DAC)  |  PWM: PC6, PC7 (TIM3)

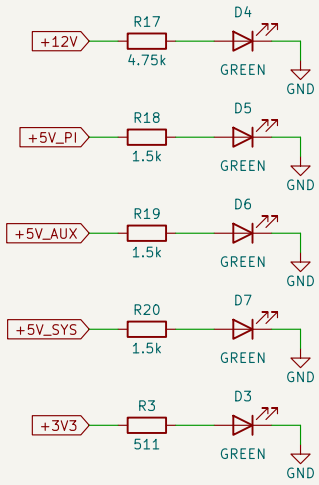
J8

Expansion header

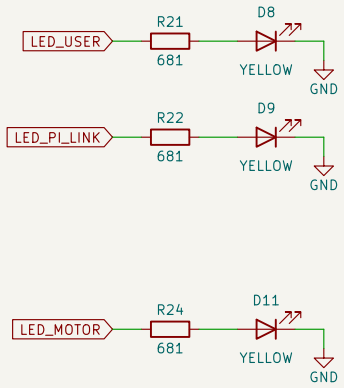



| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /MCU GPIO Breakout/ | | |
| File: GPIO_Breakout.kicad_sch | | |
| ***Title: STM32 GPIO breakout headers*** | | |
| Size: A4 | Date: 2026-10-07 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 7/13 |

Power rail indicators — one green LED per rail
(Lite-On LTST-C190KGKT, 0603, Vf 2.0 V typ)




Debug LEDs — STM32 GPIO, active high (Lite-On LTST-C190KSKT yellow, 0603)




PA5 LED_USER: heartbeat
PB12 LED_PI_LINK: Pi UART activity
PB14 LED_MOTOR: motor running / fault
681R: (3.3 − 2.1) / 681 = 1.8 mA per pin

LED current ~2–2.7 mA: R = (Vrail − 2.0 V) / I. +12V uses 4.75k for the 10.8–14.6 V battery range.
All resistors 0805; worst-case dissipation 34 mW (R17 at 14.6 V).

**Senior Design**

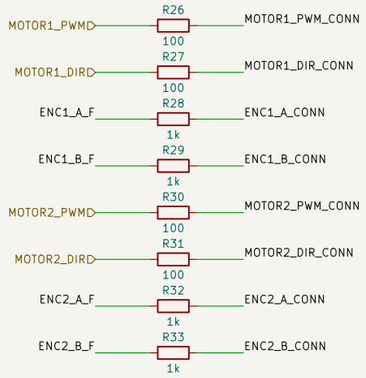
Sheet: /Power Indicators/
File: Power_Indicators.kicad_sch

***Title: Power rail and debug indicator LEDs***

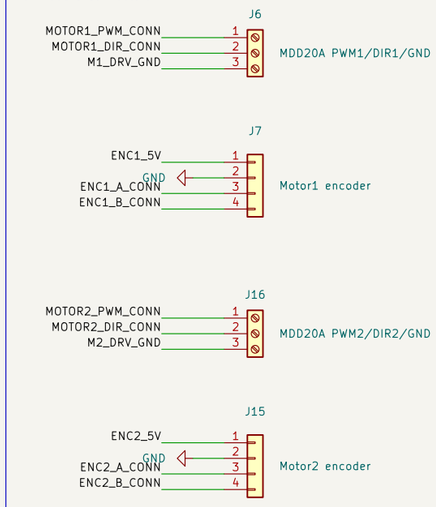
| Size: A4 | Date: 2026-10-07 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 8/13 |

Motor control — Cytron MDD20A dual-channel driver + two AndyMark PG27 motors with am-3651 encoders

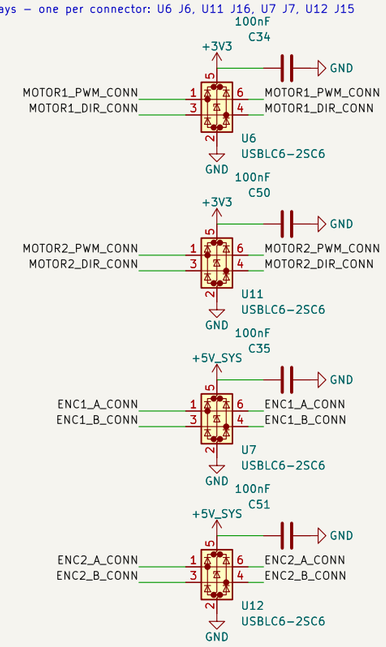
Series resistors (MCU side): 100R driver lines, 1k encoder lines



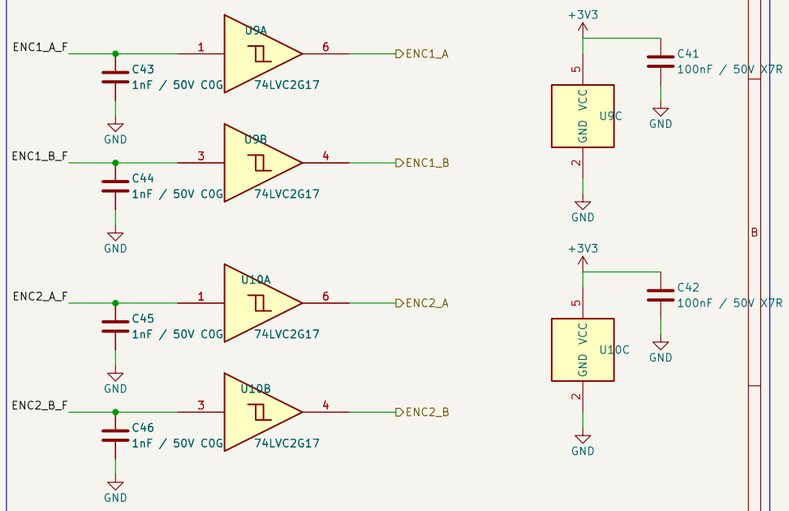

Off-board connectors



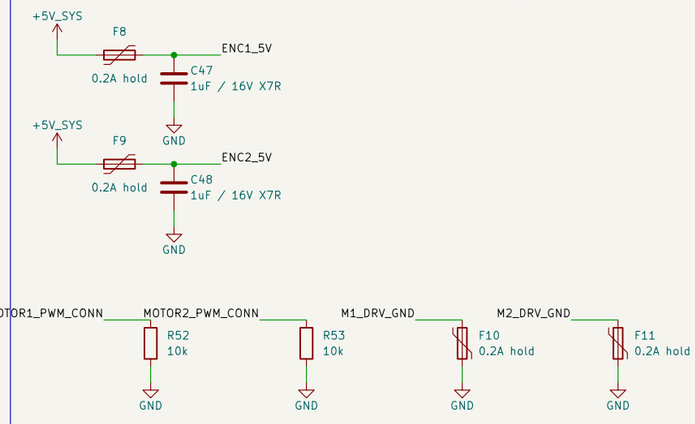

ESD arrays — one per connector: U6 J6, U11 J16, U7 J7, U12 J15




Encoder input buffers — 74LVC2G17 on +3V3 (inputs tolerate 5.5 V, even unpowered)




Encoder supply and motor-driver line protection
(If the am-3651 outputs turn out to be open-collector, bodge 4.7k from each A/B line to its ENCn_5V.)



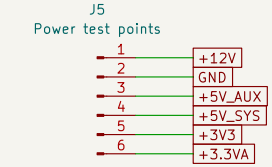

STM32F091 PIN ASSIGNMENT

| | | | | |
|---|---|---|---|---|
| MOTOR1_PWM | PA6 | TIM3_CH1 | R26 100R | J6 pin 1 -> MDD20A PWM1 (R52 10k pull-down) |
| MOTOR1_DIR | PB0 | GPIO | R27 100R | J6 pin 2 -> MDD20A DIR1 |
| ENC1_A | PA15 | TIM2_CH1 | U9 <- R28 1k | J7 pin 3 <- encoder A |
| ENC1_B | PB3 | TIM2_CH2 | U9 <- R29 1k | J7 pin 4 <- encoder B |
| MOTOR2_PWM | PA7 | TIM3_CH2 | R30 100R | J16 pin 1 -> MDD20A PWM2 (R53 10k pull-down) |
| MOTOR2_DIR | PB2 | GPIO | R31 100R | J16 pin 2 -> MDD20A DIR2 |
| ENC2_A | PA8 | TIM1_CH1 | U10 <- R32 1k | J15 pin 3 <- encoder A |
| ENC2_B | PA9 | TIM1_CH2 | U10 <- R33 1k | J15 pin 4 <- encoder B |

Timers: TIM3 = both PWMs (<= 20 kHz for MDD20A); TIM2 = encoder 1, TIM1 = encoder 2.
Encoders: 5 V parts. The +3V3 rail comes up after +5V_SYS, and ST allows only 3.6 V on any pin while VDD = 0 (AN4899), so A/B go through U9/U10 (74LVC2G17 on +3V3) instead of straight to the MCU. Never enable internal pull-ups on these pins.
PWM pull-downs: MDD20A brakes on PWM low, so a resetting MCU always means 'brake'.
J6/J16 pin 3 is a signal reference only (through F10/F11). Battery and motor leads wire directly to the MDD20A.

| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /Motor Control/ | | |
| File: Motor_Control.kicad_sch | | |
| ***Title: Motor control — MDD20A driver and encoders*** | | |
| Size: A3 | Date: 2026-10-08 | **Rev: 1.1** |
| KiCad E.D.A. 10.0.6 | | Id: 9/13 |

Power test points



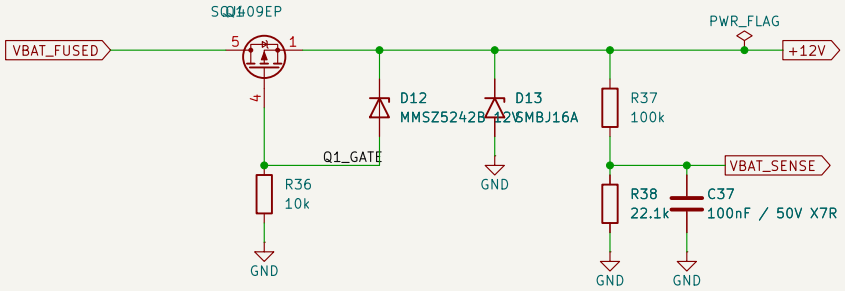

J5: one 2.54 mm header for every rail
1 +12V (after Q1)
2 GND
3 +5V_AUX
4 +5V_SYS
5 +3V3
6 +3.3VA (MCU analog)
+12V is next to GND only.




TP3: GND loop for the scope ground clip.
+5V_PI is on TP1 (Pi 5 Interface sheet),
used in the Pi bring-up check.

| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /Test Points/<br>File: Test_Points.kicad_sch | | |
| ***Title: Power test points*** | | |
| Size: A4 | Date: 2026-10-08 | **Rev: 1.1** |
| KiCad E.D.A. 10.0.6 | | Id: 10/13 |

Input protection — reverse-polarity P-FET, gate clamp, TVS, battery sense



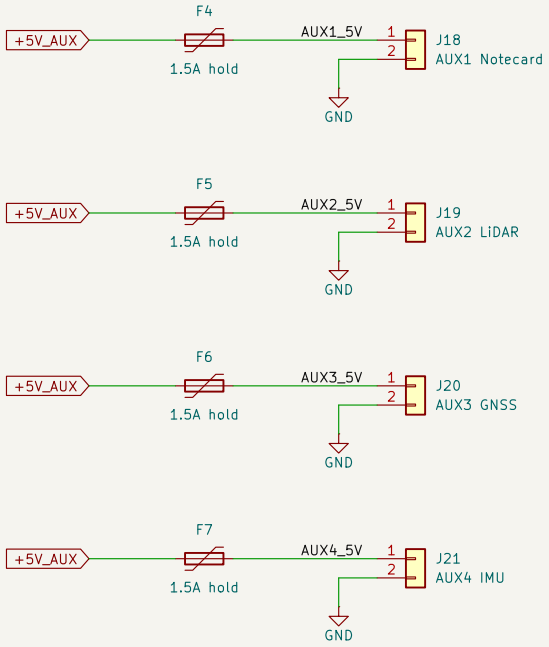

Battery chain (root sheet): J2 XT30 -> VBAT_IN -> F1 10 A MINI fuse -> VBAT_FUSED -> this sheet -> +12V.
C36 (220 uF bulk, root sheet) is on +12V, after Q1, so a reversed battery can never reverse-bias it.

Q1: high-side P-FET. A reversed battery leaves the body diode reverse-biased and VGS = 0, so nothing downstream of Q1 ever sees negative voltage. Forward: about 7 mOhm, 0.2 W at 5 A.
D12 limits |VGS| to 12 V (part limit 20 V); R36 holds the gate at GND.
D13 (SMBJ16A) clamps fast spikes on +12V to about 26 V. It is NOT sized for motor regen if the BMS disconnects (about 20 J); that must be clamped on the motor-driver side. If D13 fails short, F1 opens.

VBAT_SENSE -> PA0 (ADC_IN0): VBAT = V(PA0) x 122.1 / 22.1 (5.525 x). 10.0 V -> 1.81 V, 14.6 V -> 2.64 V.
Layout: Q1, D12, D13 between F1 and C36; D13 next to C36 with short, wide GND.

| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /Input Protection/ | | |
| File: Input_Protection.kicad_sch | | |
| ***Title: Input protection and battery sense*** | | |
| Size: A4 | Date: 2026-10-15 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 11/13 |

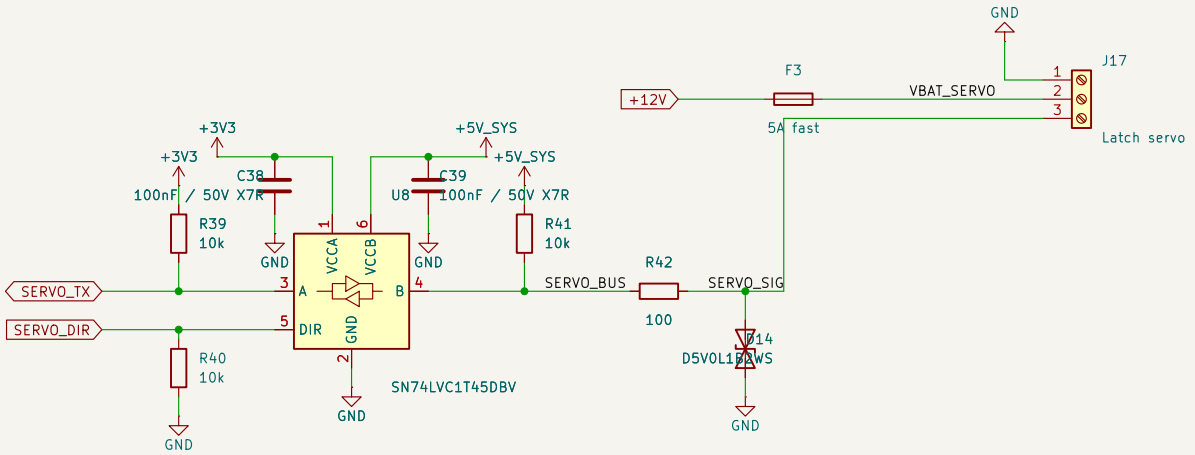
## +5V_AUX peripheral outputs — four ports, one PTC each




+5V_AUX comes from U5 (TPSM63606, 6 A), which is enabled when the Pi 5 is up (AUX_EN, see Buck 5V AUX).
Each port has its own PTC so a pinched or shorted peripheral cable only drops that port.
Connectors: JST XH 2.5 mm top-entry (plug in from above), keyed, 3 A per contact.
Use pre-made 2-pin XH cables; solder or splice the free end at each peripheral. Pin 1 = +5 V, pin 2 = GND.
PTC hold is 1.5 A on every port (trip about 3 A) so one part covers all four; bursts of 2 A for
milliseconds (Notecard modem) do not trip it. Check each peripheral's peak current before power-up.
Layout: ports on one board edge next to U5; PTCs at the connectors; +5V_AUX on a pour.

| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /AUX Outputs/ | | |
| File: AUX_Outputs.kicad_sch | | |
| ***Title: +5V_AUX peripheral outputs*** | | |
| Size: A4 | Date: 2026-10-15 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 12/13 |

Latch servo interface — Hiwonder HTS-30HS serial bus servo




HTS-30HS: 9–12.6 V (rated 11.1 V), 100 mA no-load, 3.2 A stall, half-duplex UART 115200 baud, packets start 0x55 0x55.
Power: a full 4S LiFePO4 is 14.6 V, above the servo's 12.6 V maximum, so J17 pin 2 feeds an EXTERNAL buck module set to 11.1–12 V (rated 5 A or more). Its output and GND go to the servo plug; SERVO_SIG goes straight to the plug.
Do not also tie the buck or servo ground to the battery or chassis: the servo current must return only through J17 pin 1.

Signal: PB10 = USART3_TX (AF4) in single-wire half-duplex mode (HDSEL), configured open-drain with R39 pull-up.
PB11 = SERVO_DIR: drive high just before sending, low as soon as the last byte's TC flag sets (the servo answers read commands immediately). U8 translates 3.3 V <-> 5 V; Hiwonder's reference bus has a 10k pull-up to 5 V (R41).
Firmware note: in half-duplex mode the USART also receives its own transmitted bytes; discard them.

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|---|---|---|
| **Senior Design** | | |
| Sheet: /Servo Interface/ | | |
| File: Servo_Interface.kicad_sch | | |
| ***Title: Latch servo interface (Hiwonder HTS-30HS)*** | | |
| Size: A4 | Date: 2026-10-15 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 13/13 |